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ORIGINAL ARTICLE

Integrating higher-order relations for enhanced twitter bot detection

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Abstract
Bot detection is a critical task in preserving the integrity of social networks and mitigating online disinformation. Despite advances in graph-based methods for detecting bots, these models often rely on follow relations, assuming that users sharing similar characteristics are more likely to connect. However, these methods often fail to address complex behavioral patterns indicative of coordinated bot activities, allowing bot developers to avoid detection more easily. We investigate the potential of integrating behavioral and higher-order relations, focusing on Retweet, Co-Retweet (where two users retweet the same tweet) and Co-Hashtag (where two users frequently use the same hashtag), and compare these against conventional follower and following relations. We further explore the process of relation creation, particularly highlighting the Co-Hashtag relation's robustness against data collection flaws and its ability to mitigate shortcomings in the collection of datasets. Our experiments not only contribute to the ongoing efforts to address challenges in bot detection, but also open up new avenues for exploring how complex network patterns can be employed for social network analysis.

Keywords Bot detection · Graph neural network · Relation enhancement

1 Introduction

Platforms such as X, previously known as Twitter, are now fundamental to our social interactions. They have transformed our methods of online communication, influenced public debates, and opened up channels to the most current news and viewpoints. A significant challenge within these social platforms is the widespread occurrence of bot accounts, which are capable of influencing public opinion, particularly in sensitive sectors such as politics and financial markets (Cresci 2020). Therefore, it is essential to detect these bots to preserve the integrity of social networks and fight against online misinformation. Moreover, it is notoriously hard to estimate the true extent of the presence of bots on social media platforms, and platforms may be incentivized to misrepresent them, as it could negatively impact revenue.[1] In 2017, (Varol et al. 2017) estimated that bots make up to 15% of all Twitter accounts (Varol et al. 2017). In another study, Cresci et al. analyzed Twitter discussions concerning the US stock market and concluded that up to 71% of the engaged users might have been bots (Cresci et al. 2019). Nonetheless, comparing such studies is problematic due to the varying methodologies and definitions applied in bot detection (Varol 2023).

As time progresses, bots are becoming increasingly sophisticated, a development often termed as *bot evolution* (Cresci 2020; Feng et al. 2022). This term describes the adversarial cycle in which newer bots evade increasingly more sophisticated bot detection measures by becoming progressively indistinguishable from real humans. A notable demonstration of this trend comes from a study by Cresci et al. in early 2017 (2017), where participants could only identify new bots with 24% accuracy, significantly lower than the 91% accuracy for older bots. Cresci (2020) points out that bot detection methods must be able to distinguish between genuine users and bots, which disguise themselves

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[1] https://storage.courtlistener.com/recap/gov.uscourts.cand.330648/gov.uscourts.cand.330648.257.0.pdf.

as genuine users through stolen profile pictures and neutral messages. This challenge has been further intensified by the advancement of artificial intelligence (AI), particularly generative AI, which makes it more difficult to separate individual bot accounts from genuine users. The growing challenge of differentiating AI-created text from human-written text underscores the complexity of this issue. Notably, this is highlighted by OpenAI's decision to disable their AI classifier as of July 2023 due to the low accuracy rate in distinguishing between AI-generated and human-generated content, which would result in a high number of false positives.[2]

In response to these challenges with traditional detection methods that often rely on specific account features, graph-based methods are emerging as an alternative, due to their proven effectiveness in recognizing coordinated, synchronized activities (Feng et al. 2022). By leveraging these techniques, it is not only possible to study how users interact with content, but also how they interact with other users. The rationale behind these approaches stems from the assumption that human-guided and authentic activities typically display more variability than their automated, inauthentic counterparts. This emphasizes the need to move beyond analyzing individual accounts to focusing on patterns of suspicious coordination within groups.

While graph-based methods have advanced for detecting Twitter bots, these models frequently make some homophily assumptions, implying that users sharing similar characteristics are more likely to connect. Although this holds for certain botnets, such as the recent fox8 botnet (Yang and Menczer 2023), this general assumption can lead to an oversimplification of the profiles of interconnected users. An observation that aligns with the research by Elmas et al. (2022) on retweet bots. Analysis of data acquired from black market sites revealed inconsistencies in typical assumptions regarding bot characteristics, including aspects like *volume of activity*, *diversity*, *following and followers*, and *temporality*. It was shown that bots could originate from compromised accounts, operating as bots only temporarily, with no instances found of one bot following another. Therefore, while some assumptions may be applicable to certain bot types, they do not universally hold across a broader range of bots. This, combined with recent advancements in AI, underscores the challenge of defining bots through specific metrics such as tweet volume or follower count. Fundamentally, a bot is an automated account that simulates human-like behavior, making it increasingly difficult to detect by traditional methods.

Such observations should prompt researchers to critically assess, whether the metrics used to evaluate the performance of bot detection methods are, in fact, contributing to improving downstream applications. Hays et al. (2023) argued that this is currently not the case for X/Twitter bot detection tools, attributing high performance to simplistic collection and labeling practices of the datasets employed. Furthermore, Martini et al.(2021) highlighted that bot detection can produce widely varying outcomes depending on the methodology used. This implies that current tools may not be ready for practical application and may result in the misclassification of many users (Rauchfleisch and Kaiser 2020).

With the heightened difficulty in identifying individual bot accounts, we focus our efforts on group activities and their coordinated behavior patterns. Our research aligns with research trends that focus more on the actions and behavior of groups of accounts rather than on the classification of individual accounts (Cinelli et al. 2022). This method adopts a more proactive strategy, crucial in a domain that has traditionally been reactive (Cresci et al. 2018). We explore new relations that are difficult to manipulate; any interference could significantly hinder the effectiveness of systematic automated behaviors by limiting their typical operational patterns. This builds upon previous work (Reiche et al. 2023), that demonstrated the feasibility of using coordination patterns for the purpose of bot detection, while also considering both the inherent complexities and restrictions in data availability. Acknowledging these challenges, we differentiate between first-order behavior-based relations, such as retweets (a user sharing a tweet), and higher-order relations like co-retweet (two users retweet the same tweet) and co-hashtag (two users tweet the same hashtag more than a certain number of times). The former underscores direct user actions, whereas the latter exposes common interests or themes, revealing more nuanced group activities. Considering that social botnets are typically commanded by a singular botmaster (Zhang et al. 2016), identifying coordinated behaviors is a promising step in detecting these networks. This approach is set against the current conventional method of utilizing follow relations, which are more static. We incorporate these new relations in `BotRGCN`, which utilizes multi-model user information, such as numerical and categorical properties and encoded tweet information, combined with graph structures, to enhance bot detection accuracy. Moreoever, this allows us to study and leverage X/Twitter datasets in a different way, which is of increasing importance given that access to new data is severely limited since free data access for researchers was terminated by X/Twitter in 2023, which in turn limits researchers' ability to study and train novel classifiers (Assenmacher et al. 2022; Cresci et al. 2023). Employing the same dataset and graph neural network architecture in our studies, we perform a comparative analysis between traditional follow relations and those based on behavioral patterns to evaluate their effectiveness in bot detection, thus avoiding the introduction

[2] https://openai.com/blog/new-ai-classifier-for-indicating-ai-written-text.

Table 1 Statistics (above) and in-depth analysis (below) of human and bot characteristics in TwiBot-22

Measurement	Human	Bot	Total
Users (all)	860,057	139,943	1,000,000
Users (min. 1 tweet)	818,613	115,259	933,872
Tweet	81,250,102	6,967,355	88,217,457
Following	1,038,302	78,353	1,116,655
Followers	2,383,574	243,405	2,626,979
Retweet	1,482,911	97,732	1,580,643
Hashtags	56,353,776	9,646,857	66,000,633
Measurement	Human	Bot	Diff.
Mean tweet count*	99.25	60.45	– 48.59%
Median tweet count*	56.00	40.00	– 33.33%
Mean following count†	200.22	170.21	– 16.20%
Mean follower count†	124.59	59.40	– 70.86%
Mean retweet count*	200.39	104.98	– 62.49%
Ratio: tweet / retweet	54.79	71.29	+26.17%
Ratio: hashtag / tweet	0.69	1.38	+66.66%

*users with at least 1 tweet. †users with at least 1 follower / following

of uncertainties through changes in algorithms or datasets. Although our findings did not exceed those of traditional methods, they are still competitive in accuracy, affirming the potential of this approach.

This article extends and improves on our prior work (Reiche et al. 2023) enhancing it through more in-depth analysis and additional experiments. We further explore the process of relation creation, particularly highlighting the co-hashtag relation's robustness against data collection flaws and its ability to mitigate shortcomings in the collection of datasets. Our revised evaluation now includes precision and recall metrics in addition to accuracy and the F1 score, offering a more thorough assessment of each relation. Furthermore, a generalization study demonstrates that the Co-Hashtag relation maintains strong performance across various subcommunities, underscoring its robust generalization capabilities. The new discussion and related work section briefly expands on coordination analysis and implications of restricted data availability. The paper concludes by summarizing these findings and discussing their potential to shape future research in bot detection.

2 Methodology

2.1 Dataset

We utilize the TwiBot-22 dataset for our experiments. Compared to previous datasets, TwiBot-22 includes a wider and more diverse range of relations. For an in-depth exploration of the conceptual framework of the dataset, we refer the reader to the work of Feng et al.(2022) that introduced TwiBot-22. Previous bot detection methods were restricted to rely only on Follower/Following relations between user entities and an implicit relation between users and their tweets. The TwiBot-22 dataset encompasses 14 different types of relations. In this work, we leverage the Retweet (tweet A retweets tweet B), Post (user A posts tweet B), and Discuss (tweet A discusses hashtag B) relations to derive new higher-order relations, which we then evaluate in comparison to Follower (user A is followed by user B), Following (user A follows user B). We believe that such a range of relations offers a lot of potential for the future development of more sophisticated and accurate bot detection methods. The accessibility of these diverse relations not only enhances our analytical capabilities, but also allows us to reveal hidden connections between users, cross-referencing entities, in ways previously unattainable. We refer the reader to Table 1 for an overview of TwiBot-22, comprising both statistics and an exploration of some of the characteristics that differentiate humans from bots. The upper part of the table offers a quantitative summary of the dataset, while the lower part provides a detailed examination of the differences in behavior between humans and bots. Notable distinctions include variations in tweet and follower/following counts[3] and ratios such as hashtag-to-tweet, revealing differences between the two types of accounts. This comparative analysis offers a valuable insight that guides the process of

[3] Interestingly, the total numbers of followers and following do not match, a result of the data collection methodology.

Table 2 Architecture of the BotRGCN model

BotRGCN architecture		
Input layers	Description embedding	Linear ($\mathbb{R}^{D_s \times \frac{D}{4}}$) + LeakyReLU
	Tweet embedding	linear ($\mathbb{R}^{T_s \times \frac{D}{4}}$) + LeakyReLU
	Numerical properties embedding	linear ($\mathbb{R}^{N_s \times \frac{D}{4}}$) + LeakyReLU
	Categorical properties embedding	linear ($\mathbb{R}^{C_s \times \frac{D}{4}}$) + LeakyReLU
Hidden layers	Input transformation	Linear ($\mathbb{R}^{D \times D}$) + LeakyReLU
	RGCN 1st layer	RGCN convolution ($\mathbb{R}^{D \times D}$)
	RGCN 2nd layer	RGCN convolution ($\mathbb{R}^{D \times D}$)
	Hidden transformation	Linear ($\mathbb{R}^{D \times D}$) + LeakyReLU
Output layer	Final output	linear ($\mathbb{R}^{D \times 2}$)

Variables: D: embedding size, D_s: description size, T_s: tweet size, N_s: numerical properties size, C_s: categorical properties size. The input layers' outputs are concatenated before processing through the hidden layers. A dropout regularization technique is applied between the RGCN layers. The model is used with the CrossEntropyLoss, which implicitly includes a Softmax activation on the output.

deriving new relations. We explore these aspects in more detail in the subsequent section.

Annotations for TwiBot-22 were created using a weak supervision learning strategy. Expert-labeled data from 1,000 users was used as a training set, while noisy labels for the entire dataset were generated using hand-crafted labeling functions and several bot detection models. These noisy labels were then refined through majority voting and further processed using the Snorkel framework (Ratner et al. 2017). An evaluation of this process yielded an accuracy of 90.5% on the expert-annotated test set.

In their publication introducing the TwiBot-22 dataset, Feng et al.(2022) demonstrated in an extensive benchmark experiment, comparing 35 baseline methods across 9 different datasets, that graph-based methods are generally more effective than feature-based or text-based methods, particularly with newer dataset. Notably, they also observed that the baseline methods performed worse on newer datasets compared to older ones, which could be attributed to the evolving nature of bots. From these 35 baseline methods, the best performing one was BotRGCN, which we used for our experiments.

2.2 BotRGCN

BotRGCN (Bot Detection using Relational Graph Convolutional Networks) (Feng et al. 2021) employs a graph-based approach for identifying bots on Twitter. The model first creates a multimodal representation by jointly encoding multiple numerical and categorical user properties, along with user tweets and descriptions through a pre-trained RoBERTa model (Liu et al. 2019). These encodings serve to represent individual users, capturing various aspects of their behavior and characteristics. It constructs a heterogeneous graph by establishing multiple relational neighborhoods for each user in the dataset. BotRGCN utilizes relational graph convolutional networks (RGCN) (Schlichtkrull et al. 2018), which are capable of handling diverse numbers of relations, thus enabling the model to capture complex patterns of interactions between users. The choice to utilize BotRGCN was influenced by its modular and efficiently designed structure that facilitates straightforward modifications and experimental applications. The model was initialized with hyperparameters as specified in the original setup, which can be found at the designated Github repository.[4] Additionally, adjustments were made to accommodate the specific number of categorical and numerical properties in TwiBot-22, setting the description size (D_s) and tweet size (T_s) to 768, the size of numerical properties (N_s) to 5, categorical properties size (C_s) to 3, and the embedding size (D) to 128. Detailed descriptions of the architecture and specific elements of BotRGCN are further elaborated in Table 2.

2.3 Derived relations

In their research on retweet bots, Elmas et al. (2022) highlight the complexity of identifying bots due to their non-obvious characteristics. For example, their analysis revealed that the majority of bot accounts in their dataset had more followers than the accounts they were following, and that no two bots followed each other. The researchers also observed diverse retweet patterns in terms of timing and frequency among the bots. This insight, coupled with the observation of bot evolution, led us to investigate the potential offered by new sets of relations, inspired by work from Vargas et al. (2020), which builds upon coordination patterns from Keller et al. (2020).

[4] https://github.com/BunsenFeng/BotRGCN.

We concentrate on the Retweet, Co-Retweet, and Co-Hashtag relations. These relations are behavior-based, making them inherently harder to manipulate than traditional Follower and Following relations. We believe that this approach has the potential to reveal additional patterns of coordinated behavior among users. However, none of these relations are readily usable for us out-of-the-box and require some data transformation steps.

Retweet: *a user retweeted the tweet of another user.* Our analysis showed that bots tend to retweet disproportionately. To take advantage of this, we first need to transform the existing Retweet relation from tweet→tweet to user→user. By cross-referencing the given Retweet relation with the Post relation (user→tweet), we are able to associate a user for each tweet and subsequently derive the Retweet relation in the form of user→user. See Fig. 1 for more details.

Co-Retweet: *two users retweeted the same tweet.* We introduce this relation to emphasize instances where two users retweeted the same tweet. To achieve this, we map a user to each tweet that retweets another tweet, similar to the process laid out in Retweet above. Then, we group these users by their retweeted target tweet. From these groups, we create all possible combinations of users (excluding pairs

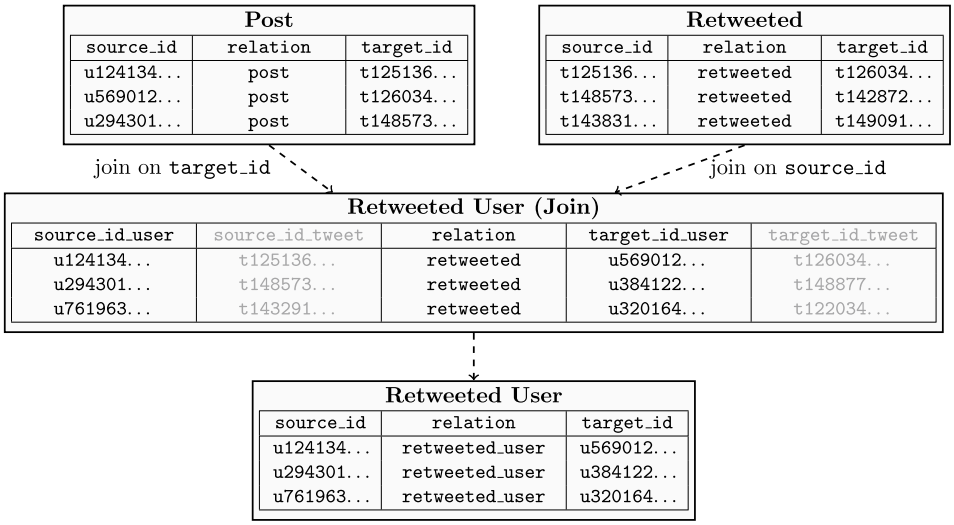


Fig. 1 Visualization of the process of deriving the new Retweet (`retweeted_user`) relation. By first splitting up the initial edges file into single relations (not shown), we then joined the `post` and `retweeted` relations, such that we get the user-id to each tweet in the Retweeted table. Lastly, we discarded the tweet ids to get the expected format

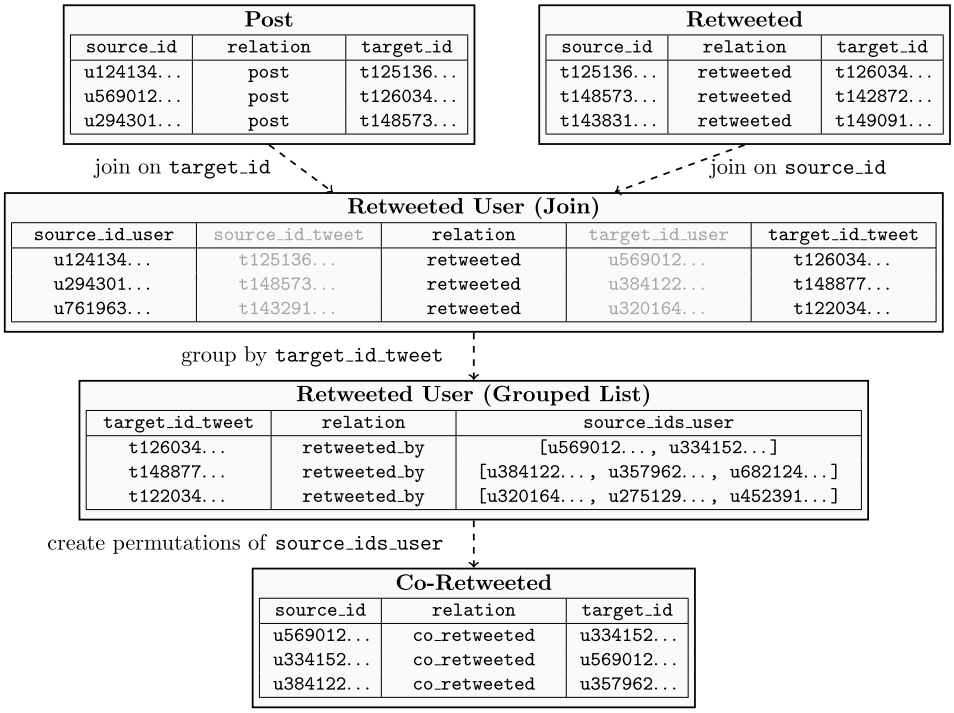


Fig. 2 Visualization of the process of deriving the new Co-Retweet (`co_retweeted`) relation. The process to obtain the Retweeted User (Join) table is similar to the process described in Fig. 1. The table is then grouped by `target_id_tweet` column. For this relation we are only interested in tweets that get retweeted by more than one user, therefore, all rows in which the the resulting list does not contain at least 2 users were discarded

with the same user twice) and export them as our new relation CO-RETWEET. This process is illustrated in Fig. 2.

CO-HASHTAG: *two users tweet the same hashtag above a certain threshold.* Using a similar grouping and pairing approach as with the CO-RETWEET relation, we focus on the DISCUSS relation (tweet→hashtag). Prior to the pairing step, we filter out hashtags with an unusually large number of users to decrease computational demands and filter out those hashtags that do not offer any reasonable insight. After this step, we create pairs of users who tweeted the same hashtag a minimum of n times. The choice of n can be regarded as a hyperparameter itself and is further detailed in the following sections of experiments and Table 4. Refer to Fig. 3 for an illustration of this process.

2.4 Limiting factors

While `TwiBot-22` is assumed to contain high-quality labels, it is important to recognize that we cannot entirely dismiss the possibility of underlying biases towards older notions of bot characteristics. A potential bias could be introduced by the use of non-transparent hand-crafted labeling functions and dependence on existing bot detection methods. Some of the employed methods were trained on follow relations, an assumption we challenged in the introduction, but more importantly, could lead to feature leakage, where follow relations get overstated in their influence. In addition, recent evidence indicates that classifiers that perform exceptionally well within one dataset may significantly underperform when applied to others, even when employing more sophisticated models (Hays et al. 2023). This may be attributed to the reliance on inherently unstable features present in the initial training data.

These observations might, in fact, result from the inherent limitations in the collection and the labeling practices of the employed datasets. This is troublesome, given that the insights we can get about the field are almost entirely limited to information we can obtain from such datasets. An implication of this is that the datasets used for bot detection are not widely representative and instead are significantly shaped by the sampling and labeling processes involved. We can observe that similar shortcomings are also present in `TwiBot-22`, as can be seen in Table 3. Therefore, although `TwiBot-22`'s expert-guide process signals a marked improvement, the broader methodology might compromise the overall effectiveness of the dataset. However, this should not solely be attributed to the creators of the dataset, but also considered as a consequence of the very nature of social botnet dataset creation (Assenmacher et al. 2022; Cresci et al. 2023). Nevertheless, we assume the labels in the data to be the ground truth, due to the lack of better annotation methods and the complex nature of this issue.

Originating from the collection around seed users, only a small subset of users serve as source nodes in the unidirectional graph, stemming from the FOLLOWER and FOLLOWING relation. The method of gathering data is very similar

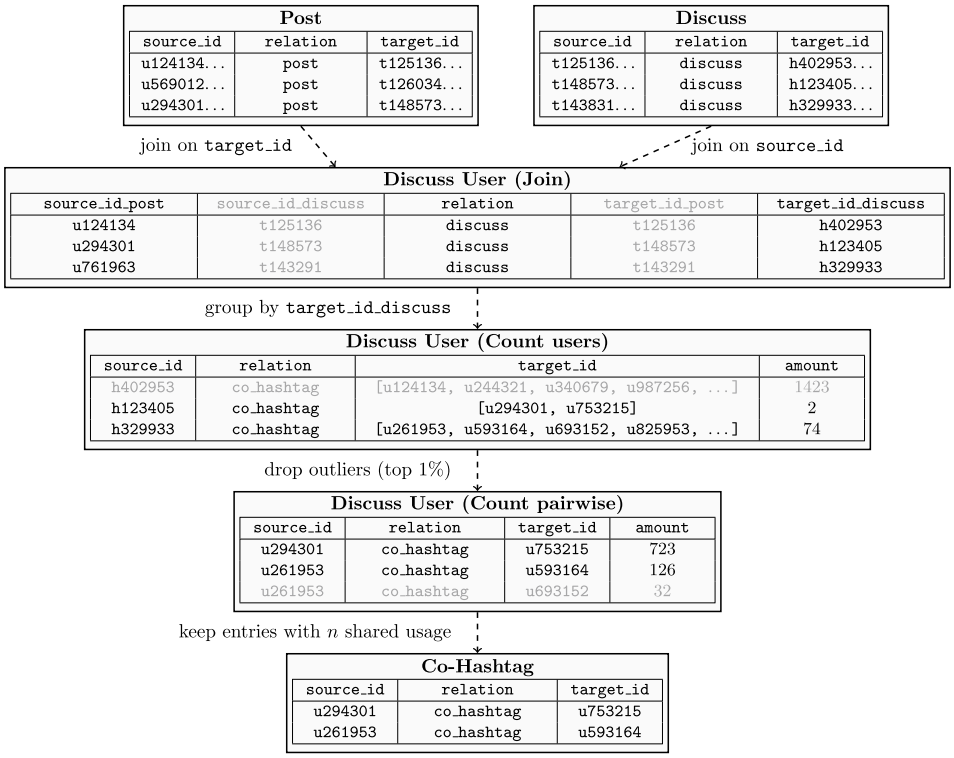


Fig. 3 Visualization of the process of deriving the new CO-HASHTAG (`co_hashtag`) relation. We join the `post` and `discuss` relation to associate user-ids with each hashtag in the `discuss` table. In this example, we assume a threshold `amount` value of 100, below which `co_hashtag` occurrences are discarded. We then create pairs of users with the respective count of how often they share a hashtag. Lastly, we keep only those with at least n shared hashtags and discard the `amount` column to obtain the expected format

Table 3 Comparison of the amount of distinct values for the `source_id` and `target_id` columns of the FOLLOWER and FOLLOWING relations in the `TwiBot-22` dataset

Relation	Distinct `source_id`		Distinct `target_id`	
	Human	Bot	Human	Bot
FOLLOWER	8,334 (0.97%)	1,319 (0.94%)	370,265 (43.05%)	60,748 (43.41%)
FOLLOWING	8,506 (0.99%)	1,430 (1.02%)	483,169 (56.18%)	44,518 (31.81%)
RETWEETED	7,400 (0.86%)	931 (0.66%)	216,800 (25.21%)	12,310 (8.80%)
CO-RETWEETED	5,652 (0.66%)	598 (0.43%)	5,652 (0.66%)	598 (0.43%)
CO-HASHTAG	497,681 (57.87%)	64,080 (45.79%)	497,681 (57.87%)	64,080 (45.79%)

It reveals a notable discrepancy in the way relations are concentrated among users. The percentage values in parentheses following each number represent the proportion of distinct users covered by their respective relation and category, based on a total of 860,057 human and 139,943 bot accounts

Table 4 Sensitivity study of the CO-HASHTAG edge creation threshold

Threshold	Amount	Accuracy (%)	F1-score (%)
3	2,340,420	74.58	36.61
4	3,780,800	74.95	40.17
5	5,326,662	75.07	40.31
6	6,954,818	75.25	43.39
19	32,396,274	75.24	41.36

The Amount column corresponds to the parameter n, representing the minimum number of times pairs of users tweeted the same hashtag. We run each experiment five times and report the average value

to that used in TwiBot-20 (Feng et al. 2021), where the network is built around seed users through a breadth-first traversal. However, it is worth noting that the TwiBot-20 dataset has been subject to criticism by Hays et al.(2023). Consequently, most users are not connected to the graph, a sharp contrasts with the requirements for effectively employing graph neural networks. Unfortunately, this is also the case for the RETWEETED and CO-RETWEETED relations, since they are directly derived from the provided RETWEET (tweet→ tweet) relations of the dataset. It should be noted at this point that a sparser graph is expected for the retweet relations, as similar observations have been made for the fox8 botnet (Yang and Menczer 2023). In contrast, the CO-HASHTAG relation allows us to reach up to 75.66% of all users in the dataset if all edges are kept. Furthermore, due to its bidirectional nature, it avoids the issue of clustering around seed users, as detailed in Table 3.

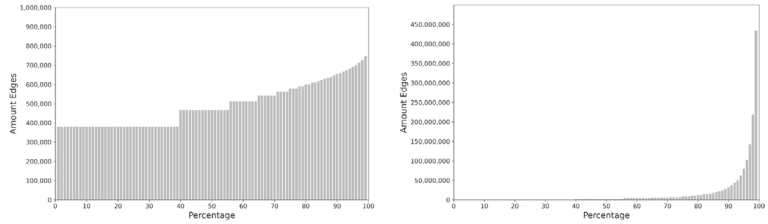
Consequently, the use of the CO-HASHTAG relation enables a comprehensive analysis of the communication patterns and structural dynamics of users in the dataset, providing valuable information that overcomes some of the inherent limitations of the sampling procedure.

3 Experiments

To determine the feasibility of utilizing coordination patterns for bot detection, we conducted sensitivity and ablation studies, maintaining constant hyperparameters across all experiments. The model initialization followed the specifications described in the `BotRGCN` subsection, with a dropout rate of 0.3, a learning rate of 0.001, and a weight decay of 0.005. We utilized the train/test split from `TwiBot-22` to ensure comparability with prior work. In contrast to the experiment setup in Reiche et al. (2023), we increased the number of training epochs to 1000. This produced more stable and better results, improving the reliability of our evaluations.

3.1 Comparing thresholds in relation creation

To find the best threshold for the CO-HASHTAG relation we aim for a Pareto-optimal point with high coverage of nodes while keeping the computational load, i.e. the amount of edges, manageable. To achieve this, we adjusted the percentile values of hashtag co-occurrence frequencies within our dataset to determine the effects on edge counts across



(a) Amount Users (b) Amount Edges

Fig. 4 Amount user (Fig. 4a) and amount edges (Fig. 4b) depending on the percentage cutoff for the CO-HASHTAG relation. The numbers supporting these graphs can be found in Table 9 in the appendix

various threshold settings. Our examination of Fig. 4a and b reveals that significant changes in node coverage occurred at percentile cutoffs of 41, 56, 65, and 71, which correspond to threshold values of 3, 4, 5, and 6, respectively. Moreover, the analysis of Fig. 4b identified a key knee point at the 90% cutoff, matching a threshold of 19. However, our results from Table 4 indicated that the threshold of 6, corresponding to the 71% percentile, offered the most effective results.

Consequently, we have established a threshold of 6 for the Co-Hashtag relation in our ongoing analyses, unless otherwise noted. This setting balances computational demands with comprehensive graph coverage.

3.2 Comparing relations and ablation study

To thoroughly evaluate the impact of different relational configurations on our bot detection model, we performed a sensitivity analysis. The results of these tests, which focus on accuracy, F1-score, precision, and recall, are detailed in Table 5. This provides a foundation for comparing these configurations and assessing performance improvements from newly derived behavioral relations.

Notably, the combination of Follower+Following relations achieved the highest accuracy, confirming the expectation that these are effective for bot detection. This matches the intuition that these relations can be a strong indicator. However, our main interest was the newly derived behavioral relations, with the *follow* relations serving as a baseline for comparison.

Our findings necessitate contextual interpretation, contrasting our approach with the conventional use of Follower+Following relations. Instead of relying solely on these foundational indicators, our approach explores behavioral patterns to detect more complex interactions, such as mutual retweeting dynamics or consistent use of the same hashtags beyond certain thresholds. While challenges related to data collection have limited our use of retweet-based relations, we focused our further experiments on the Co-Hashtag relation. However, we do not dismiss the Retweet relation and still consider it valuable for future exploration (Table 5).

Although we did not outperform the conventional approach, our results using only the Co-Hashtag are closely competitive, with differences of 1.85 percent points lower in accuracy and 4.93 percent points in F1-score. This gap, although initially discouraging, reveals upon closer examination the capability to make predictions, avoiding biases that might have characterized previous approaches, as outlined in Limiting Factors subsection. Additionally, integrating the Retweeted relation into our analysis significantly enhanced the recall of our predictions, effectively increasing our model's capability to identify all actual bots. This improvement is particularly noteworthy because it was achieved without expanding user coverage, as documented in Table 3, but rather by introducing an alternative detection signal that captured bots missed by relying solely on the Follower and Following relations. Nevertheless, this improvement in recall resulted in a decrease in precision. This means that the model was somewhat more likely to misclassify legitimate users as bots, potentially leading to a higher count of false positives. However, this is not a desirable property of bot detection as even a small rate of false positives could lead to a high number of human users being wrongly classified as bots and artificially increase the perceived number of bots in follow-up tasks (Rauchfleisch and Kaiser 2020).

Expanding on the findings from Feng et al. (2021), we have carried out an ablation study of `BotRGCN`, utilizing the same layer configuration. Irrespective of the chosen relations, this ablation study reaffirms that bot detection requires a multifaceted approach, integrating various

Table 5 Sensitivity analysis of `BotRGCN` to different edge types in the graph

Category	Sensitivy settings	Accuracy (%)	F1-score (%)	Precision (%)	Recall (%)
Single relation	`follower`	78.70	57.34	70.42	49.10
	`following`	77.21	46.37	75.51	34.10
	`retweeted`	77.02	49.03	70.75	37.81
	`co-retweeted`	76.99	50.30	69.09	39.75
	`co-hashtag`	77.22	51.60	68.99	41.29
Two relations	`follower+following`	79.07	56.53	72.98	46.40
	`retweeted+co-retweeted`	76.52	46.16	71.22	34.62
	`co-retweeted+co-hashtag`	77.19	50.58	70.00	40.24
Three relations	`following+follower+retweeted`	78.77	58.69	68.75	51.27
	`following+follower+co-hashtag`	78.73	55.81	72.37	45.78
	`retweeted+co-retweeted+co-hashtag`	76.95	50.02	69.39	39.45
Five relations	all of the above	78.67	56.49	70.84	47.08

We run each experiment five times and report the average value

modalities. This approach must then model the aggregation of these signals, aiming to ensure a clear distinction between accounts involved in automated coordinated efforts and those demonstrating authentic behavior, which may stem from social initiatives. The results of our experiments, detailed in Table 6, highlight the critical role of this multifaceted approach in enhancing the accuracy and reliability of our detection algorithms.

These findings underscore the potential benefits of a multirelational approach, although it is important to consider that characteristics inherent to the dataset might affect these observations. By integrating all available numercial and categorical user properties alongside encoded user tweets, `BotRGCN` addresses the challenge of identifying bots on an individual level. Additionally, by utilizing behavioral patterns to construct a graph, it addresses the issue of bots, that appear genuine individually but act in coordination. The combination of these elements results in a more comprehensive and nuanced detection strategy, and bot developers may find it challenging to avoid behavior-based detection without substantially constraining their capabilities.

3.3 Generalization study

To ensure effective generalization of bot detection algorithms to unseen data, it is crucial to recognize that achieving high scores on common machine learning metrics does not automatically imply robust generalization capabilities (Echeverría et al. 2018). Motivated by this, we initiated a generalization study focusing on the Co-Hashtag relation, using ten sub-communities as identified in the `TwiBot-22` dataset.

The creation of these sub-communities involved a two-part strategy: initially, five subcommunities anchored around specific influential users from varied interest domains, such as politics, technology, and entertainment were selected. Subsequently, clustering was performed using K-means on the word2vec (Mikolov et al. 2013) representations of hashtags used in their tweets, further organizing users into sub-communities based on the similarity of their hashtag usage. Using these sub-communities, we conducted cross-validation by training the model on one community (i) and evaluating it on another (j), as depicted in Fig. 5. This approach allowed us to assess how well the

Table 6 Ablation Study of `BotRGCN` under different relation types using 2 layers of `RGCN`

Ablation setting	Accuracy (%)	F1-score (%)	Precision (%)	Recall (%)
Follower + following				
`RGCN` + T	70.51	2.04	46.56	1.05
`RGCN` + T, N	72.38	24.82	62.63	15.68
`RGCN` + T, N, C	74.11	36.08	66.27	24.83
`RGCN` + T, N, C, D (`BotRGCN`)	79.07	56.53	72.98	46.40
Co-hashtag				
`RGCN` + T	70.58	3.35	51.91	1.74
`RGCN` + T, N	71.62	17.90	60.82	10.81
`RGCN` + T, N, C	73.46	29.94	67.39	19.41
`RGCN` + T, N, C, D (`BotRGCN`)	77.22	51.60	68.99	41.29

Abbreviations used: T = User Tweets; N = User Numerical Properties; C = User Categorical Properties; D = User Descriptions. We run each experiment five times and report the average value

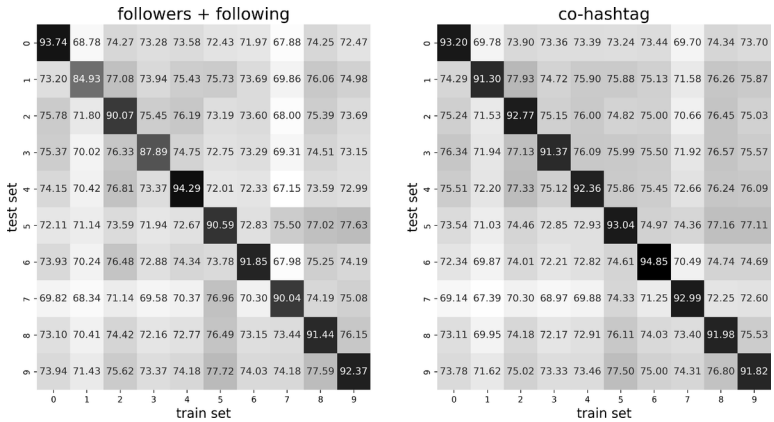


Fig. 5 Training models on community i and testing on community j, we present both model accuracy and F1 score. The report includes the average values of each heatmap, which serve as an overall indicator of the generalizability of the models, calculated over 5 runs

Co-Hashtag relation generalizes compared to the established Followers+Following baseline.

In the comparative analysis, the Co-Hashtag relation exhibits a comparably strong, if not marginally better, performance against the Followers+Following baseline. This close performance is particularly significant given the potential biases in data labeling and collection we discussed earlier. While we cannot definitely confirm whether the biases exist or have skewed the results, the Co-Hashtag relation consistently shows strong performance. That said, even if we set these assumptions aside, our findings still indicate a potentially enhanced generalization capability inherent to the Co-Hashtag relation, suggesting that utilizating more complex user interactions could be beneficial in improving the adaptability and effectiveness of bot detection across varying data contexts.

3.4 Evaluation on TwiBot-20

In this section, we explore whether the Co-Hashtag relation can also be applied to other graph-based bot detection datasets, specifically testing its applicability on `TwiBot-20` (Feng et al. 2021). This dataset shares some structural similarities with `TwiBot-22`, as both were created by expanding along follow edges using a breadth-first search. However, it captures a different point in time, contains considerably fewer users, and was annotated through a traditional crowdsourcing process. In contrast to `TwiBot-22`'s 90.5% annotation accuracy, TwiBot-20 achieves 80% (Feng et al. 2022). Given these characteristics, we aim to verify whether the Co-Hashtag relation performs similarly in this context (Table 7).

Table 7 Sensitivity study of the Co-Hashtag edge creation threshold for `TwiBot-20`

Threshold	Amount	Accuracy (%)	F1-score (%)
3	4.741.572	84.93	85.80
4	4.713.198	85.81	87.03
5	4.684.818	86.28	87.61
6	4.657.378	85.74	86.41
7	4.629.358	84.95	86.74

Similar to Table 4, the Amount column corresponds to the parameter n, representing the minimum number of times pairs of users tweeted the same hashtag. We run each experiment three times and report the average value

However, due to the limited relational data in `TwiBot-20`, we are unable to directly apply the same Co-Hashtag derivation approach described in earlier sections. Specifically, `TwiBot-20` lacks the Discuss and Post relations that were used in the process outlined in subsection 2.3. As a result, we instead attempt to extract hashtag usage directly from tweet texts. This approach has several drawbacks compared to utilizing the relational structures, as it may lead to incomplete data due to issues like text truncation or formatting inconsistencies (e.g., caused by emojis). Nevertheless, it still provides an approximate representation of the Co-Hashtag relations.

Once the hashtags for each user were extracted, we followed the same steps outlined in Fig. 3: grouping by hashtags, filtering out outliers, and retaining only those entries where the hashtag was used by more than one user. After completing these steps, we conducted a sensitivity analysis to determine the appropriate threshold for relation creation. The results are presented in Table 8. As shown, varying the threshold does not significantly affect the number of edges in the graph. Unfortunately, the overall number of users who shared the same hashtag with at least one other user is relatively low (7,190 users), indicating that threshold filtering is less impactful for smaller datasets compared to `TwiBot-22`.

Table 8 presents the results of training `BotRGCN` using the follower + following relations and then again using the Co-Hashtag relation. The performance of both approaches is closely aligned. Notably, in this case, the Co-Hashtag relation slightly outperforms the *follow* relations. This further supports that the Co-Hashtag relation is a robust approach for bot detection, further validating its effectiveness.

4 Discussion and related work

Our presented approach centers on the detection and classification of individual bots, utilizing coordinated relations that are also widely used in the broader field of coordination analysis. For example, Retweet and Co-Retweet networks have been used to identify and characterize coordinated astroturfing (Keller et al. 2020), while others have explored Co-Tweet or Co-Reply signals to uncover coordinated behavior (Graham et al. 2024). Moreover, Jacobs et al. (2023) have shown effectiveness in detecting coordination amplification by analyzing hashtag usage. Ng and Carley (2023) introduced a combined synchronization index to measure the coordination among users. This method underscores the potential of weighted

Table 8 Comparison of results obtained from training `BotRGCN` on different edge types

Relations	Accuracy (%)	F1-score (%)	Precision (%)	Recall (%)
`follower+following`	86.02	87.55	84.50	90.83
`co-hashtag`	86.28	87.61	85.65	89.69

We run each experiment three times and report the average value

coordination metrics - an extension that could further enhance the approach discussed in our work.

In contrast to general coordination analysis, where the goal is often to understand collective behaviors or uncover coordinated campaigns, our work directly targets the identification of bots on an individual level. While similar relations, like Co-Retweet, have been employed to analyze group behavior in networks (Pacheco et al. 2021), our research applies these relations to the classification of individual accounts. These studies, along with the ones mentioned earlier primarily extract coordination signals from textual content. In contrast, our method relies on relational data. We did not compare our findings with more datasets due to the lack of similarly rich relation data. Despite the effectiveness of both approaches, their varying methodologies make a direct comparison difficult.

The challenges of limited data availability and the lack of accessible benchmark datasets has been acknowledged in recent research (Assenmacher et al. 2022). These issues have also impacted our research, highlighting the need for more datasets of similar richness, such as `TwiBot-22`, to enable broader comparisons and thorough validation of findings.

The recent restrictions on data access, particularly due to changes in X/Twitter's API policies, have greatly limited researchers' ability to conduct effective research, while bot developers remain largely unaffected (Cresci et al. 2023). Although this situation is new for X/Twitter, it reminds of earlier limitations imposed by Facebook following the Cambridge Analytica scandal. Despite this, research in the field continued, with some researchers shifting their focus to more accessible platforms, while others resorted to more fragmented data collection methods, such as web scraping (Freelon 2018). Although often necessary, this approach raises ethical issues related to data privacy (Trezza 2023).

Unfortunately, in today's environment, easily accessible social media data sources are increasingly scarce, presenting considerable challenges for ongoing research. The reality is that most social media research is highly dependent on the cooperation of platforms to share and provide access to their data (Assenmacher et al. 2022). As access to data becomes increasingly restricted, it is crucial to explore ways to maximize the potential of existing datasets.

However, there are promising efforts aimed at addressing these issues. In collaboration with OpenMined, X/Twitter is exploring new methods to provide researchers with secure and compliant access that goes beyond what is currently offered via public APIs.[5] Although these initiatives are still in the early stages, they offer some optimism for overcoming the current limitations on data access.

In the meantime, our work has leveraged data that remains available, demonstrating promising results. Although our approach was demonstrated on X/Twitter, it is adaptable to other social media platforms, providing a practical solution in a landscape where data accessibility remains a challenge.

5 Conclusion

Our research has shown that incorporating higher-order relations such as Co-Hashtag and Co-Retweet can yield competitive results compared to traditional follower relations, while also reducing biases from data collection methods. Our results underscore the effectiveness of using coordination patterns for bot identification, using subtle signs of coordinated bot behavior. Specifically, the Co-Hashtag relation provides a strong indicator for detecting bots, with high generalization capabilities across various subcommunities of X/Twitter.

Additionally, our ablation study indicates that although direct follow relations are good indicators of bot activities, augmenting these models with behavioral relations may improve their ability to detect bots. Such integration can increase sensitivity to new types of bot behaviors that might be missed by conventional approach.

The `TwiBot-22` dataset utilized in this study has played a crucial role in identifying novel relations. Our findings suggest that incorporating these relations enhances the analysis of social networks and the detection of bots. Nonetheless, extracting them is complex and can take a significant amount of time. In an effort to promote further research and ensure reproducibility, our code has been made accessible on GitHub.[6]

Despite the benefits of these complex relations, there are still challenges due to the limitations and representativeness of the datasets. With current restrictions on accessing new data, the value of existing datasets becomes even more critical. Optimizing the analytical use of available data ensures that detection mechanisms can continue to effectively counter increasingly sophisticated bot operations. Looking ahead, it is essential to continue to advance methods that can keep pace with the fast-changing dynamics of online interactions and ongoing developments in bots.

Appendix A Threshold variations for the Co-Hashtag relation

Table 9 displays the edge counts for the Co-Hashtag relation across various threshold settings, determined by the percentile ranks of hashtag co-occurrence frequencies

[5] https://blog.x.com/engineering/en_us/topics/insights/2022/investing-in-privacy-enhancing-tech-to-advance-transparency-in-ML.

[6] https://github.com/iamsreiche/twibot22-relations.

Table 9 Edge count of the Co-Hashtag edge creation threshold

Threshold (%)	Amount	Distinct users	Threshold
100	–	–	82,690
99	433,548,296	746,583	179
98	218,377,954	727,152	94
97	142,481,412	713,246	64
96	101,941,364	701,353	48
95	79,277,326	692,161	39
94	62,310,912	682,673	32
93	50,499,534	673,854	27
92	43,515,058	667,436	24
91	36,763,066	659,815	21
90	32,396,274	653,857	19
89	28,111,626	647,076	17
88	23,890,210	638,872	15
87	21,846,154	634,173	14
86	19,854,436	629,041	13
85	17,899,860	623,319	12
84	15,981,694	616,645	11
82–83	14,093,912	609,222	10
80–81	12,233,262	600,418	9
78–79	10,409,428	590,066	8
75–77	8,652,544	577,589	7
71–74	6,954,818	561,761	6
65–70	5,326,662	541,242	5
56–64	3,780,800	512,162	4
40–55	2,340,420	467,331	3
1–39	1,047,456	380,026	2

within our dataset. This approach concentrates our analysis on the most significant and actively used hashtag pairs. The table specifies the number of distinct users for each threshold, highlighting how changes in these thresholds influence both the graph sparsity in our dataset.

Author contributions S.R. origination of research idea, carried out experiments, analyzed results and wrote the main manuscript text.S.C. and K.S. provided critical feedback, helped with the analysis, helped to refine the paper.T.F. oversaw project direction.

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Data availability The experiments in this manuscript where carried out using the `TwiBot-22` dataset, which can be requested with an institutional email address as explained here:https://github.com/LuoUndergradXJTU/`TwiBot-22`

Declarations

Conflict of interest The authors declare no competing interests.

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